

Abstract :

This project report describes the design, construction, and working principle of a Line Following Robot without using any microcontroller. The system is fully hardware-based and operates using IR sensors, comparator circuits, and motor driver ICs. The main objective of this project is to understand the fundamental concepts of robotics, sensor interfacing, and decision-making using analog and digital electronics. This project is suitable for beginners and academic purposes, especially for understanding automation without software dependency.

Table of Contents:

Page No

Abstract :	01
List of Figures :	01
List of Tables :	01
1. Introduction :	01
2. Working and Design of Circuit:	01-02
▪ Description of Components	03
▪ Mathematical Formula :	03-04
3. Simulation :	05
4. Project Implementation :	05
5. Future Improvement :	05-06
6. Conclusion :	06

List of Figures :

Figure 1: Circuit Diagram of Line Following Robot without Microcontroller.

Figure 2: Simulation.

Figure 3: Project Implementation.

List of Tables :

Table 1: Truth Table of Robot Movement

1. Introduction :

A line following robot is an autonomous robotic system capable of detecting and following a predefined line on the floor. The line is usually black on a white surface or vice versa. In this project, the robot is designed without using any microcontroller, which makes the system simple, cost-effective, and easy to understand. The robot uses IR sensors to detect the line and comparator circuits to process sensor signals. Based on the sensor output, the motor driver controls the movement of the robot.

2. Working and Design of Circuit:

➤ Working :

The working principle of the line following robot is based on infrared (IR) sensing technology. IR sensors continuously sense the surface beneath the robot. When the sensor detects a black line, the reflected IR light is less compared to a white surface. The comparator circuit converts the analog sensor signal into a digital logic level. These logic signals are fed to the motor driver IC, which drives the DC motors accordingly.

➤ **Design of Circuit :**

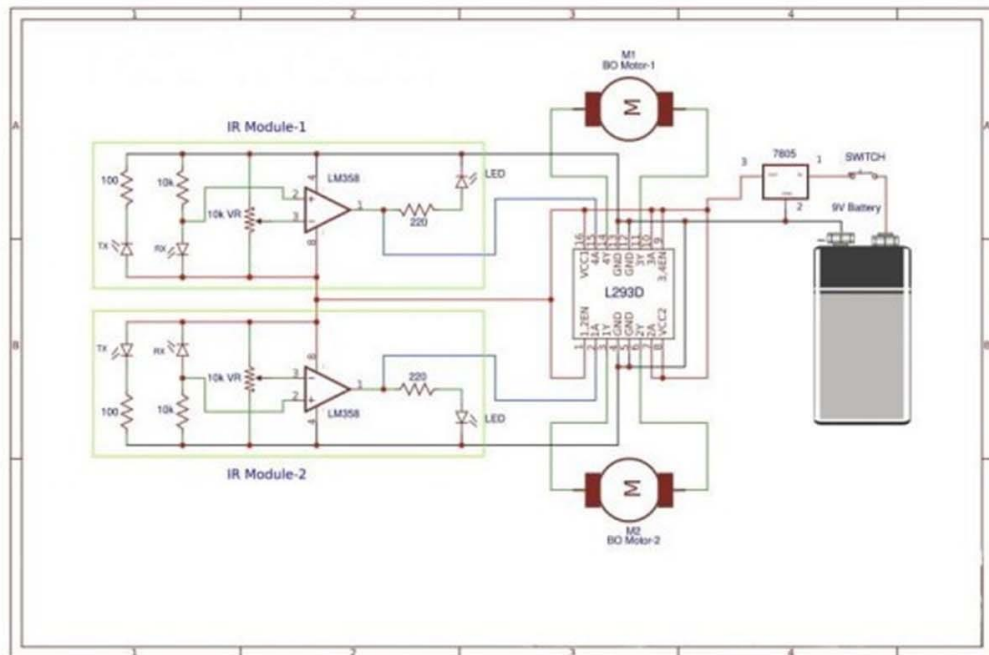


FIG 1 : Circuit Diagram of Line Following Robot without Microcontroller.

➤ **Description of Components :**

	Required Equipment • IR Proximity Sensor • L293D IC • BO Motor • Switch • 7805 Voltage Regulator • Zero P C B • Pin Header • Wires • 9V Battery • Wooden Frame (9cm x 4cm, 9cm x 2.5cm) • Wheels
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➤ **Mathematical Formula:**

1. LDR (Light Dependent Resistor) Resistance Formula

$$R_{LDR} = \frac{V_{out}}{V_{in} - V_{out}} \times R$$

Where:

R_{LDR} = LDR resistance

V_{in} = Input voltage

V_{out} = Output voltage

R = Fixed resistor value

2. Voltage Divider Formula (Sensor Output Voltage)

$$V_{out} = V_{in} \times \frac{R_{LDR}}{R + R_{LDR}}$$

Where:

V_{out} = Sensor output voltage

V_{in} = Supply voltage

R_{LDR} = LDR resistance

R = Fixed resistor

3. DC Motor Speed Formula

$$N = \frac{V - I_a R_a}{K \theta}$$

Where:

N = Motor speed (RPM)

V = Applied voltage

I_a = Armature current

R_a = Armature resistance

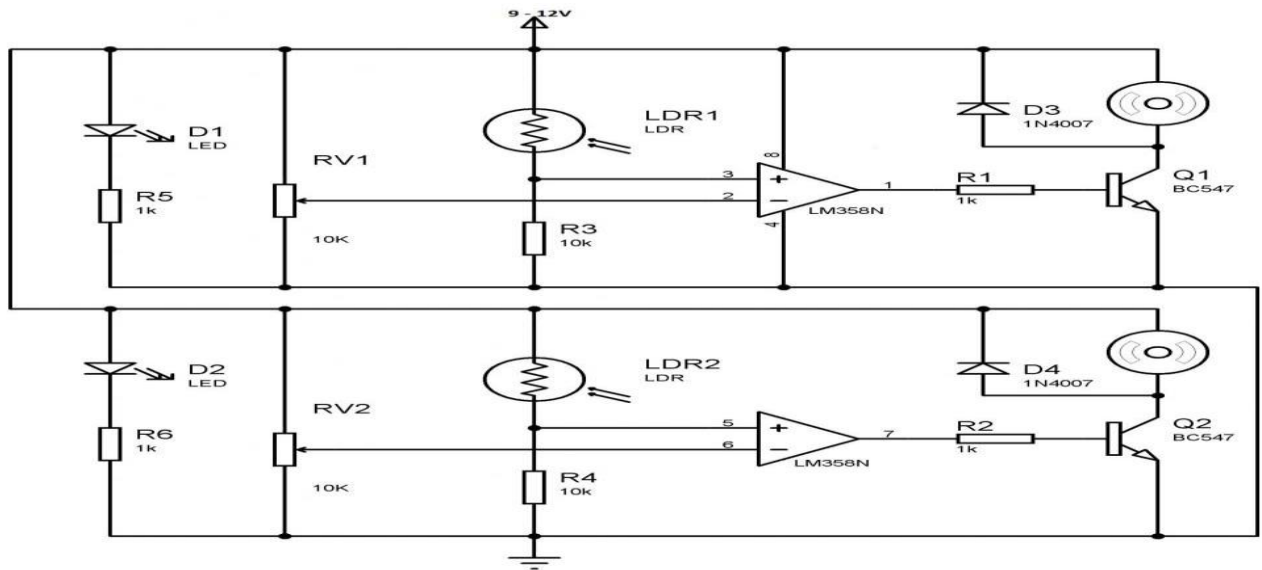
K = Motor constant

θ = Flux

Table 1: Truth Table of Robot Movement

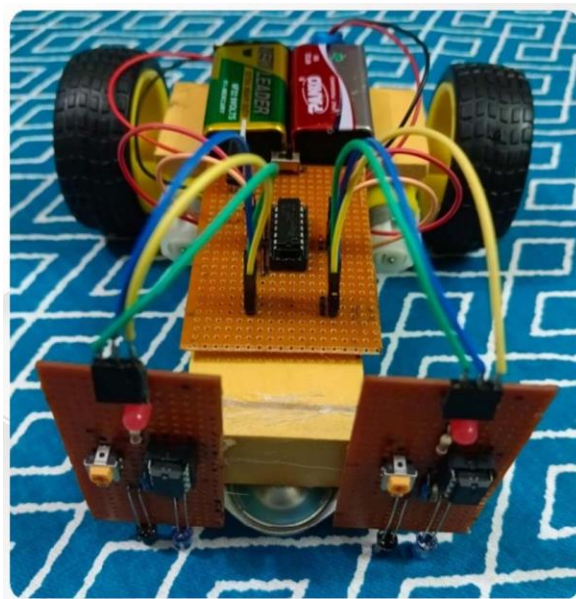
Left Sensor	Right Sensor	Left Motor	Right Motor	Robot Movement
0	0	ON	ON	Moves Forward
0	1	OFF	ON	Turns Right
1	0	ON	OFF	Turns Left
1	1	OFF	OFF	Stops

3. Simulation:



Figuer 2: Simulation.

4.Project Implementation :



Figuer 2: Project Implementation

5. Future Improvement :

Although the line following robot without microcontroller works successfully, several improvements can be made to enhance its performance and efficiency.

Use of Microcontroller

In future, a microcontroller (such as Arduino or PIC) can be used to make the robot more intelligent. This will allow better decision-making, flexible programming, and advanced control.

Improved Sensors

IR sensors can be replaced with more accurate sensors like color sensors or array sensors to detect lines more precisely, especially in varying light conditions.

Speed Control System

PWM (Pulse Width Modulation) technique can be added to control motor speed smoothly, which will improve turning accuracy and overall motion.

Obstacle Detection

Ultrasonic or IR proximity sensors can be added so that the robot can detect and avoid obstacles automatically.

Battery Management System

A rechargeable battery with voltage regulation can be used to improve power efficiency and increase operating time.

Better Mechanical Design

Using lightweight materials and better wheel alignment can increase stability, speed, and durability of the robot.

Wireless Control and Monitoring

Bluetooth or Wi-Fi modules can be added to monitor robot movement or control it remotely.

6. Conclusion :

In this project, a Line Following Robot without using any microcontroller was successfully designed and implemented. The robot is able to detect a black line on a white surface using IR sensors and follow the path using simple electronic components such as comparators, motor driver ICs, and DC motors.

This project helps to understand the basic working principle of sensors, comparators, and motor control circuits. It is cost-effective, easy to build, and suitable for beginners to learn robotics and electronics. Although the system has some limitations compared to microcontroller-based robots, it performs efficiently for simple applications.

Overall, this project is a good foundation for learning automation and robotics, and it can be further improved by adding advanced control techniques in the future.